Project INSPIRE Design Document

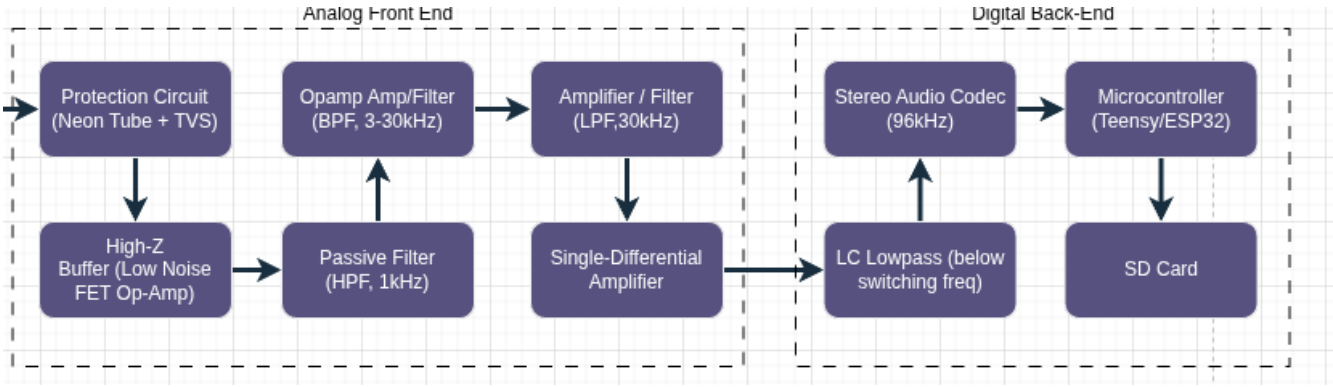
Project Part VIII: ADC

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

The Analog-Digital Converter (ADC) is the single most important component, as its performance dictates the translation of analog voltage signals into digital audio streams that can be processed and stored by the microcontroller. Because of the low amplitude of signals from the VLF antenna, the ADC must be able to establish a good dynamic range and have a clean, low noise floor output spectrum.

- Either a 24-bit or higher architecture, in order to meet or exceed the 100dB Signal-to-Noise Ratio (SNR) or dynamic range.
- The ADC must support a sampling rate of 96kHz, which ensures the 30kHz edge of the band is sampled and not aliased down into lower frequencies.
- The ADC must output data either through I2S or similar audio protocols in order to ease digital code development and work with the existing audio hardware on the microcontroller.
- The ADC must work with 3.3V digital logic that the microcontroller outputs, and must accept 3.3V analog input signals in order to ensure compatibility with the rest of the system.

• 3) Expected Outputs

The sub-circuit requires a technical write-up and hardware plan containing the following:

- **Component Selection**: Compare suitable high-resolution audio codecs or standalone ADCs, and find one suitable to meet the requirements provided, with justification.
- **Interfacing Plan:** Read the datasheet and find the necessary digital clocking system needed to interface with the audio stream of the codec/ADC. Determine the needed frequencies to operate the device at the needed sampling rate.
- **Verification Plan:** Determine a testing protocol to verify the performance of the converter, and a digital signal integrity

test to ensure no issues with the bitstreams between the ADC and the microcontroller.

    - A spectral analysis of a captured sine wave will be helpful to verify SNR, and effective number of bits.
    - Oscilloscope and/or Signal Analyzers will be helpful to read the digital bus lines and determine the signal lines are not interrupting or deteriorating the audio stream or control messages.

Optional Extra (Stretch Goals):

- If all else is completed, find an applications schematic provided by the manufacturer, and determine any other passives or other components that might be necessary for the ADC to operate.